11/09/23 AN

Hall Ticket Number:

MR-21

Code: B313CB6S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, Aug/Sep-2023

Computer Organization and Architecture

(Common to CSB, CSM & CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Give an overview of the Instruction formats.	2	3	1	2
1. b)	Explain fetch phase of instruction cycle.	2	2	1	1
1. c)	List logical Microoperations.	2	1	2	1
1. d)	Justify why indexed addressing is used for array processing.	2	5	2	4
1. e)	Write about Floating point representation with an example.	2	1	5	2
1. f)	Explain signed magnitude representation of fixed point numbers.	2	2	5	1
1. g)	What is polling?	2	3	3	1
1. h)	Distinguish between the asynchronous and synchronous data transfer.	2	1	3	2
1. i)	Distinguish between loosely coupled and tightly coupled multiprocessors.	2	4	4	1
1. j)	What is cache coherence?	2	1	4	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Draw and explain the functional blocks of a Digital computer.	5	1	1	1
2. b)	Write a short note on ALU design.	5	2	1	2
	OR				
2. c)	Define Register Transfer Language. List and explain Arithmetic Microoperations.	5	2	1	1
2. d)	Illustrate about tri-state buffer. Justify why tristate buffers are used widely in Bus system.	5	5	1	1
3. a)	Define the following (a) Microoperation b) Micro instruction (c) Microprogram (d) subroutine.	5	1	2	2
3. b)	With the help of a neat sketch explain general register organization.	5	1	2	1
	OR				
3. c)	Explain different instruction formats with examples.	5	1	2	1
3. d)	Differentiate between the direct and indirect addressing with examples.	5	4	2	1
4. a)	Illustrate the various representations of the signed numbers.	5	1	5	1
4. b)	Give the flowchart for doing decimal division and also explain the sequence of operations of it.	5	2	5	1
	OR				
4. c)	Give and explain the hardware implementation of the signed magnitude addition and subtraction.	5	1	5	3
4. d)	With the help of a block diagram explain Decimal Arithmetic Unit.	5	3	5	1
5. a)	Explain about the various cache memory techniques.	5	2	3	1
5. b)	Explain daisy chaining with a neat sketch.	5	1	3	1
	OR				
5. c)	Draw and explain DMA with a neat sketch.	5	4	3	1
5. d)	Explain in detail about the memory hierarchy in the processor.	5	1	3	1
6. a)	Distinguish between RISC and CISC processor.	5	4	4	2
6. b)	Explain about the arithmetic pipeline.	5	1	4	1
	OR				
6. c)	Explain vector processing.	5	2	4	2
6. d)	Write a short note on I/O processor.	5	1	4	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome